

Experiment 'p_3_5_order' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_5_order

Natural language statement: Write a method that takes three numbers and returns "increasing" if they are in increasing order, "decreasing" if they are in decreasing order, and "neither" otherwise. Here, "increasing" means "strictly increasing", with each value larger than its predecessor. The sequence 3 4 4 would not be considered increasing.

Method signature: p_3_5_order (i: int, j: int, k: int) returns (s: string)

Ensures

- if $i < j < k$ then $s == \text{"increasing"}$ else if $i > j \ \&\& \ j > k$ then $s == \text{"decreasing"}$ else $s == \text{"neither"}$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes three numbers and returns "increasing" if they are in
→ increasing order, "decreasing" if they are in decreasing order, and "neither"
→ otherwise. Here, "increasing" means "strictly increasing", with each value larger
→ than its predecessor. The sequence 3 4 4 would not be considered increasing.

The signature should be:

`p_3_5_order (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if i < j < k then s == "increasing" else if i > j && j > k then s == "decreasing"`
→ `else s == "neither"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dmdq8fnO0IldT2K6w3TrRQ8mmtDEg

Model: gpt-5.5-2026-04-23

Created at: 1780485260

Finish reason: stop

Usage: 446 tokens in, and 227 tokens out

```
{
  if i < j && j < k {
    s := "increasing";
  } else if i > j && j > k {
    s := "decreasing";
  } else {
    s := "neither";
  }
}
```

Final Program

```
method p_3_5_order (i: int, j: int, k: int) returns (s: string)
  ensures if i < j < k then s == "increasing" else if i > j && j > k then s == "
    → decreasing" else s == "neither"
```

```
{  
  if i < j && j < k {  
    s := "increasing";  
  } else if i > j && j > k {  
    s := "decreasing";  
  } else {  
    s := "neither";  
  }  
}
```

Total Token Usage

Input tokens: 446

Output tokens: 227

Reasoning tokens: 148

Sum of 'total tokens': 673

Experiment Timings

Overall Experiment started at 1780485260061, ended at 1780485270481, lasting 10420ms (10.42 seconds)

Iteration #1 started at 1780485260061, ended at 1780485270481, lasting 10420ms (10.42 seconds)